### Table 4.2 CloudFormation parameter properties

| Property | Description | Example |
|---|---|---|
| `Default` | A default value for the parameter | `Default: 'm5.large'` |
| `NoEcho` | Hides the parameter value in all graphical tools (useful for secrets) | `NoEcho: true` |
| `AllowedValues` | Specifies possible values for the parameter | `AllowedValues: [1, 2, 3]` |
| `AllowedPattern` | More generic than `AllowedValues` because it uses a regular expression | `AllowedPattern: '[a-zA-Z0-9]*'` allows only a–z, A–Z, and 0–9 with any length |
| `MinLength, MaxLength` | Defines how long a parameter can be | `MinLength: 12` |
| `MinValue, MaxValue` | Used in combination with the `Number` type to define lower and upper bounds | `MaxValue: 10` |
| `ConstraintDescription` | A string that explains the constraint when the constraint is violated | `ConstraintDescription: 'Maximum value is 10.'` |

A parameter section of a CloudFormation template could look like this:

```
Parameters:
   KeyName:
     Description: 'Key Pair name'
     Type: 'AWS::EC2::KeyPair::KeyName'    ①
   NumberOfVirtualMachines:
     Description: 'How many virtual machine do you like?'
     Type: Number
     Default: 1                            ②
     MinValue: 1
```